EXPERIMENT 10

Non-Contiguous Memory Allocation: FIFO

AIM:

To implement the First In First Out (FIFO) page replacement algorithm and calculate the total number of page faults.

PROCEDURE:

1. Declare the size with respect to page length

2. Check the need of replacement from the page to memory

3. Check the need of replacement from old page to new page in memory 4.

Form a queue to hold all pages

5. Insert the page require memory into the queue

6. Check for bad replacement and page fault

7. Get the number of processes to be inserted

8. Display the values

PROGRAM:

```
def fifo():
    global a, n, m
    f = -1
    page_faults = 0
    page = []

    for i in range(m):
        page.append(-1)

    for i in range(n):
        flag = 0

        for j in range(m):
            if page[j] == a[i]:
                flag = 1
                break

        if flag == 0:
            f = (f + 1) % m
            page[f] = a[i]
            page_faults += 1
```

```
            print("\n%d ->" % a[i], end=" ")

            for j in range(m):
                if page[j] != -1:
                    print(page[j], end=" ")
                else:
                    print("-", end=" ")
        else:
            print("\n%d -> No Page Fault" % a[i], end=" ")

    print("\n Total page faults : %d." % page_faults)

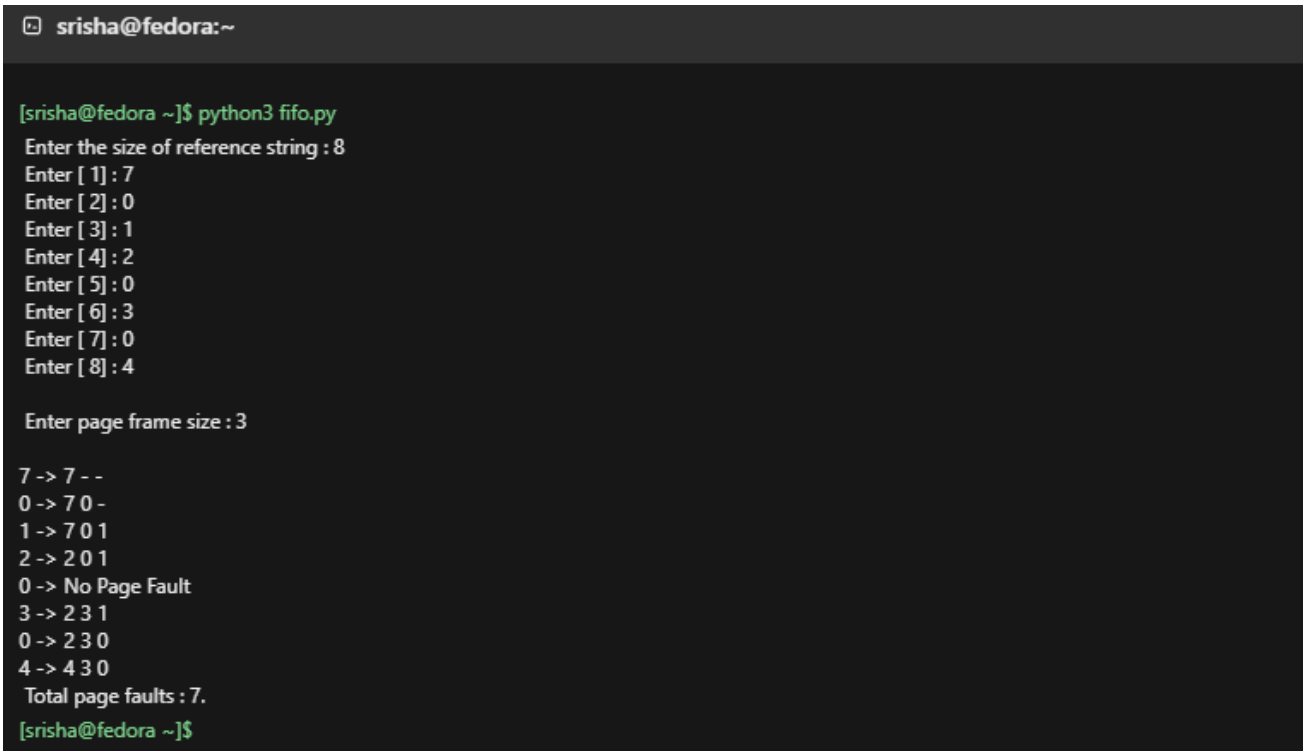

a = []
n = int(input("\n Enter the size of reference string : "))

for i in range(n):
    a.append(int(input(" Enter [%2d] : " % (i + 1))))

m = int(input("\n Enter page frame size : "))

fifo()
```

OUTPUT:



RESULT:

Thus, the First In First Out (FIFO) page replacement algorithm was implemented and executed successfully using Python in Fedora Linux, and the total number of page faults was calculated.